

Underwater Image Enhancement
via a Hybrid UResNet and Computer Vision Pipeline

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Abstract

Underwater images often suffer from low contrast, color distortion, and visibility degradation due to light absorption and scattering. This work presents a hybrid enhancement framework that combines a deep learning-based model with traditional computer vision techniques. The proposed method adopts UResNet as the core network for global feature restoration, while an optimized post-processing pipeline—comprising CLAHE, gamma correction, and saturation adjustment—is used to refine local contrast and color accuracy. Quantitative evaluation using the UCIQE metric demonstrates an average improvement of approximately 5% over the baseline UResNet model. Moreover, owing to its compact architecture and efficient computation, the proposed approach achieves **SOTA** performance on the LSUI dataset under low-cost, resource-constrained conditions. These results highlight the effectiveness and practicality of integrating learning-based and traditional enhancement techniques for real-time underwater image restoration.

Keywords: UIE, UResNet, Computer Version

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1 Introduction

Reliable underwater vision is fundamental for applications such as ROV/AUV inspection, aquaculture monitoring, and marine exploration. However, underwater imaging suffers from severe degradation caused by wavelength-dependent attenuation and backscatter. Red light is absorbed most rapidly, while scattered light adds a grayish veil that reduces local contrast and edge definition. As a result, underwater images often exhibit strong blue–green color casts, low visibility, and spatially varying illumination. These effects not only hinder human interpretation but also degrade the performance of downstream vision tasks such as detection, mapping, and SLAM.

Recent advances in deep learning have enabled significant progress in underwater image enhancement (UIE). Encoder–decoder architectures such as U-Net [1] and its derivatives (e.g., UResNet [2]) can learn complex mappings from degraded to clean images when trained on paired or synthetic datasets. Despite their success, these models face two practical limitations.

First, generalization remains weak. Weak generalization refers to a model’s limited ability to adapt to unseen conditions or data distributions that differ from those in the training set. While the network may perform well on images similar to its training data, it often fails to preserve consistency under varying lighting conditions, turbidity levels, or color distortions commonly encountered in underwater environments. This limitation arises because the model implicitly learns statistical correlations rather than true physical priors of underwater imaging. As a result, deep networks may generate visually plausible but physically inaccurate results when confronted with novel scenarios, emphasizing the need for supplementary processing or domain adaptation strategies.

Second, computational cost and retraining needs limit deployment on resource-constrained embedded platforms. Although, diffusion-based generative models have emerged as powerful image restoration frameworks, offering fine-grained control and high perceptual fidelity.

Their iterative sampling makes them considerably slower and less predictable in real-time scenarios. In our experiments, a compact diffusion-based UIE model achieved visually pleasing color balance in low-light cases but underperformed the UResNet baseline in both quantitative metrics and temporal consistency, highlighting a trade-off between quality and practicality.

Classical computer vision encompasses algorithms and image processing techniques that are explicitly designed by human experts based on physical models and perceptual heuristics. Unlike data-driven deep learning methods, these approaches—such as histogram equalization, color correction, and edge enhancement—operate through interpretable, hand-crafted operations that embody domain knowledge about light propagation, color balance, and human visual perception. Because their mechanisms are explicitly defined rather than learned, classical methods offer stability, transparency, and predictable behavior, making them valuable complements to deep learning systems that may exhibit weak generalization. When integrated, the human-engineered precision of classical vision can effectively compensate for the statistical bias of neural networks, leading to more robust and reliable image enhancement outcomes.

Motivated by these observations, this dissertation investigates how classical computer vision (CV) enhancement techniques can complement deep learning models—particularly UResNet—through lightweight post-processing. Traditional methods such as Contrast Limited Adaptive Histogram Equalization (CLAHE), gamma correction, and saturation adjustment remain computationally efficient and interpretable. When used as a tunable pipeline after a deep model’s output, they can correct color imbalance, enhance contrast, and improve objective quality without additional training or heavy computation.

The goal of this research is therefore pragmatic: to enhance underwater imagery in terms of chromatic contrast, saturation, and brightness consistency, as reflected by improvements in the UCIQE score, while maintaining a lightweight and deployable framework. The use of the compact UResNet model, combined with a traditional computer vision pipeline, enables real-time enhancement performance even on low-power devices such as Raspberry Pi. This demonstrates that substantial perceptual and practical gains can be achieved without increasing model complexity or data requirements.

1.1 Research questions

Q1. How effectively can traditional enhancement methods—such as CLAHE, gamma correction, and saturation adjustment—improve the perceptual quality of deep UIE models like UResNet?

Q2. How does a lightweight diffusion-based UIE model compare to UResNet in terms of perceptual quality, stability, and computational cost?

Q3. What combination of model architecture and classical post-processing offers the best trade-off between quantitative quality (UCIQE), visual realism, and runtime efficiency for embedded deployment?

2 Literature Review

2.1 Traditional Computer Vision Methods for Underwater Image Enhancement

Before the deep learning era, underwater image enhancement primarily relied on classical image processing techniques designed to correct color imbalance, low contrast, and illumination non-uniformity. These approaches were rooted in histogram manipulation, gamma correction, and color model transformation, offering low computational complexity and clear interpretability—attributes particularly valuable for real-time underwater systems operating on limited hardware resources.

One representative technique is adaptive gamma correction, which adjusts image luminance based on the distribution of pixel intensities. An efficient example of this class was introduced by Huang et al [3], who proposed an adaptive gamma correction framework that modifies image histograms using a weighting distribution. Their method automatically transforms dim or poorly illuminated images by optimizing the brightness curve according to the probability distribution of luminance pixels. This adaptive approach mitigates the risk of over-enhancement commonly observed in fixed gamma correction, while maintaining global brightness consistency. Furthermore, the authors extended the technique to video enhancement by leveraging temporal information between frames, significantly reducing computational overhead while preserving temporal smoothness. Experimental comparisons demonstrated that their approach could produce image quality comparable to or superior to other state-of-the-art methods, confirming the effectiveness of gamma-based tone mapping for contrast restoration.

Complementary to gamma-based enhancement, histogram equalization and its variants have been among the most widely used methods for contrast improvement in underwater imagery. However, standard histogram equalization often causes over-saturation and noise

amplification in regions of uniform brightness. To address these issues, Tan et al. introduced the Mixture Contrast Limited Adaptive Histogram Equalization (Mixture CLAHE) method[4], a refinement of the well-known CLAHE algorithm. Their method applies CLAHE separately in both RGB and HSV color spaces to exploit complementary color information, and then fuses the enhanced outputs using the Euclidean norm. This multi-space operation helps balance local contrast enhancement with color stability, producing results that maintain natural visual appearance while avoiding common side effects such as halo artifacts or excessive noise. The experimental evaluation—conducted on images captured from Redang Island and Bidong Island—showed notable improvements in visibility, edge definition, and overall aesthetic quality. Compared to conventional CLAHE, the mixture approach effectively reduces speckle noise and mitigates over-enhancement, demonstrating better adaptability to complex underwater lighting variations.

Together, these studies illustrate the strengths and limitations of traditional contrast enhancement strategies. Methods based on adaptive gamma correction and CLAHE remain attractive due to their interpretability, low computational cost, and robustness in scenarios where learning-based models cannot be deployed. However, they typically operate on global or local statistical assumptions without explicitly modeling the physical imaging process underwater. As a result, while they improve perceptual clarity, they cannot fully recover true color fidelity or spatial consistency under severe turbidity.

In summary, traditional computer vision-based techniques established an important foundation for underwater image enhancement by emphasizing efficiency and interpretability. Yet, their inability to capture nonlinear degradation patterns and color distortions under diverse conditions motivated the transition toward data-driven methods. The next stage of research, therefore, explored convolutional neural networks (CNNs) such as U-Net and its variants, which learn complex mappings between degraded and reference images directly from data.

2.2 CNN-Based Methods for Underwater Image Enhancement

2.2.1 Overview of CNN-Based Enhancement

Convolutional Neural Networks (CNNs) have profoundly reshaped the landscape of underwater image enhancement over the past decade. CNNs provide a data-driven framework capable of learning complex

mappings directly from degraded underwater images to their enhanced versions. Through multiple layers of convolutions, nonlinear activations, and pooling operations, CNNs can hierarchically extract spatial and semantic features that represent both local textures and global illumination trends. These characteristics make CNNs particularly effective for modeling the nonlinear distortions induced by light absorption and scattering in water.

However, the underwater imaging environment poses unique challenges to CNN design. Unlike natural scenes, underwater images exhibit severe color shifts—often toward bluish or greenish tones—due to wavelength-dependent light attenuation. Scattering by suspended particles further reduces image contrast and sharpness. These degradations vary dynamically with depth, turbidity, and illumination conditions, meaning that the model must learn not only general image priors but also the complex physics of underwater optics. Consequently, CNN-based methods for underwater enhancement have evolved from simple feedforward structures into more sophisticated architectures, often integrating multi-scale learning, attention mechanisms, and domain adaptation to improve robustness and generalization.

A fundamental direction in this evolution has been the exploration of encoder–decoder frameworks, particularly the U-Net [1] family. The U-shaped structure provides a balance between feature abstraction and spatial fidelity, which is essential for restoring fine details in underwater scenes. As researchers began to integrate residual learning, attention modules, and hybrid convolutional operations into U-Net [1] variants, the CNN paradigm for underwater image enhancement transitioned from straightforward reconstruction networks to specialized architectures tailored for color restoration, contrast enhancement, and real-time underwater perception.

Lightweight Autoencoder-Based Networks

A parallel line of research focuses on the computational efficiency of underwater enhancement networks. While advanced CNN and Transformer-based methods offer superior performance, they often demand high computational resources, making them impractical for real-time applications on embedded underwater platforms such as autonomous underwater vehicles (AUVs) or remotely operated vehicles (ROVs). To bridge this gap, lightweight CNN-based autoencoders like AquaAE [6] have been introduced.

AquaAE employs a simplified encoder–decoder design with skip connections to fuse encoder and decoder features, similar to a reduced U-Net structure. Its innovation lies in its red-channel enhancement mechanism, which compensates for the spectral attenuation of longer wavelengths (red light) in underwater environments. By dynamically adjusting enhancement coefficients based on water type and depth conditions, AquaAE restores color balance while maintaining low computational complexity. In addition, the model relies solely on basic convolution and upsampling operations, avoiding the use of computationally expensive attention or transformer blocks.

The performance of AquaAE highlights an important trade-off in underwater enhancement: while deeper or more complex models achieve higher accuracy, lightweight architectures can achieve real-time performance with minimal accuracy degradation. Evaluations on benchmark datasets such as EUVP and Underwater ImageNet demonstrate that AquaAE achieves competitive PSNR and SSIM scores while requiring only a fraction of the floating-point operations and parameters compared to more complex networks. This makes it particularly suitable for embedded systems and power-constrained underwater platforms.

2.2.2 ResNet and Its Influence on Underwater Enhancement

The introduction of Residual Networks [1] (ResNet) marked a major paradigm shift in the design of deep convolutional architectures. Prior to ResNet, the depth of CNNs was constrained by vanishing gradients and optimization instability, which limited the representational capacity of enhancement models. ResNet overcame these limitations by reformulating the learning process into residual mappings—instead of learning a direct function $H(x)$, the network learns the residual

$$F(x) = H(x) - x$$

$F(x) = H(x) - x$, which is then added back to the input through skip connections. This simple yet powerful design allowed networks to exceed depths of 100 layers while maintaining stable gradients and high accuracy.

For underwater image enhancement, the residual learning concept is particularly appealing. The degradation between a clean and a distorted underwater image can itself be viewed as a residual component—a set of color and contrast distortions that, when estimated and added back to the input, yield the restored image. This analogy makes ResNet architectures naturally suited for enhancement tasks, as they learn to predict the “correction signal” required to invert underwater degradation.

Furthermore, the identity mapping inherent in ResNet preserves low-frequency information, ensuring that the enhanced image retains the natural structure and texture of the original scene.

2.2.3 Expanding the Residual Paradigm: ResNeXt and SE-Net

While ResNet introduced depth as a new dimension of learning, subsequent developments explored how to extend the representational power of residual blocks without excessively increasing computational cost. ResNeXt [7] introduced the concept of cardinality, representing the number of parallel transformations aggregated within each residual block. This design allows multiple lightweight feature transformations to be learned simultaneously, providing a form of implicit ensemble learning inside a single network layer. For underwater enhancement, where degradation patterns vary spatially across the image, high cardinality enables the model to capture diverse correction functions tailored to different local conditions.

Another influential development was the Squeeze-and-Excitation Network (SE-Net)[8], which introduced an adaptive mechanism for recalibrating feature channels. Instead of treating all channels equally, SE blocks explicitly model inter-channel dependencies through a two-step process: a global squeeze operation that captures contextual statistics, followed by an excitation operation that generates per-channel weights to emphasize informative features and suppress redundant ones. This mechanism has been particularly effective in underwater imaging, where the importance of color channels varies dramatically due to wavelength-dependent absorption. By assigning higher weights to attenuated channels (e.g., red) and lower weights to dominant channels (e.g., blue-green), SE modules help restore balanced color representations.

Integrating SE blocks into residual frameworks provides both theoretical and practical advantages. The channel-wise recalibration complements the residual connections' ability to preserve structure, yielding models that are both deeper and more discriminative. Moreover, the computational overhead of SE blocks is minimal, making them well-suited for real-time underwater enhancement. This synergy between depth (ResNet), cardinality (ResNeXt), and channel attention (SE-Net) forms the architectural foundation for many modern enhancement networks, including UResNet and its variants.

The Deep Residual Learning framework also paved the way for transfer learning in underwater vision. Networks pretrained on large-scale

natural image datasets such as ImageNet could be fine-tuned for underwater domains with relatively small amounts of labeled data. This approach significantly reduced training cost and improved convergence stability, especially when paired datasets were scarce. ResNet-based enhancement pipelines commonly replace standard encoder backbones with residual blocks to capture both local and contextual cues in underwater scenes. These architectures have been shown to outperform earlier U-Net variants, particularly in challenging conditions with severe color loss and light attenuation.

2.2.4 Comparative Analysis of CNN-Based Enhancement Frameworks

When examining the progression from conventional CNNs to UResNet, several evolutionary trends become apparent. First, multi-scale feature representation has become a central design principle. The introduction of input-level fusion modules in semi-supervised U-Net models and the hierarchical aggregation in ResNeXt both highlight the importance of capturing features across spatial resolutions. Multi-scale design enables models to simultaneously handle global illumination gradients and local texture restoration.

Second, attention and adaptive weighting mechanisms have been increasingly integrated into CNN backbones. Channel-wise recalibration (as in SE blocks) and spatial attention modules allow networks to selectively focus on degraded regions [8], leading to more efficient learning and visually consistent results. This adaptivity is particularly beneficial for underwater scenes with nonuniform lighting and spatially varying turbidity.

Third, model efficiency and real-time adaptability have emerged as critical considerations. Lightweight architectures such as AquaAE demonstrate that significant reductions in parameter count and computational complexity are achievable without a severe drop in enhancement quality. This is essential for practical deployment on embedded underwater platforms that operate under power and bandwidth constraints. As underwater robotics and real-time monitoring applications expand, efficient CNNs will remain an important research direction.

Fourth, integration of physical priors with deep learning has shown marked benefits. Hybrid frameworks that incorporate underwater optical models—either explicitly in preprocessing or implicitly through loss functions—tend to exhibit better generalization and stability. The

UResNet-VCSE [9] pipeline exemplifies how such synergy can produce visually pleasing and physically consistent results.

Finally, semi-supervised and unpaired learning paradigms address the fundamental data scarcity problem. The reliance on large-scale paired datasets has historically limited the applicability of supervised CNNs. Semi-supervised and CycleGAN-based data augmentation strategies allow the training of deeper models without extensive labeled data, thus broadening the usability of CNN-based methods in diverse marine conditions.

2.2.5 Emerging Challenges and Future Perspectives

Despite substantial progress, CNN-based underwater image enhancement still faces several challenges. The first lies in domain generalization. Networks trained on one dataset or water type often fail to generalize to others due to differences in light absorption spectra and particle composition. While semi-supervised methods alleviate this to some extent, the lack of domain-invariant feature learning remains a barrier. Future research may explore contrastive learning or domain adaptation techniques to improve robustness.

The second challenge involves balancing model complexity with interpretability. Deeper and more parameter-rich models achieve better restoration quality but often act as black boxes, making it difficult to interpret how enhancement decisions are made. Incorporating explicit physical priors or explainable intermediate representations, as in UResNet with hybrid preprocessing, provides a promising route toward interpretable deep enhancement.

Third, temporal consistency in underwater video enhancement remains underexplored. Most CNN-based methods operate on still images, leading to flickering when applied frame by frame. Integrating temporal coherence constraints or recurrent architectures could yield smoother and more stable video outputs suitable for real-time underwater navigation and surveillance.

Lastly, while CNNs remain powerful, recent advances in diffusion models and transformer-based architectures suggest a new direction for underwater image enhancement. These models can capture long-range dependencies and probabilistic representations of uncertainty, offering potential improvements in realism and generalization beyond what traditional convolutional frameworks can achieve.

2.2.6 CNN Method Summary

In summary, CNN-based methods have formed the backbone of underwater image enhancement research. Starting from classical convolutional architectures, the field has progressed through increasingly sophisticated designs—U-Net introduced structural preservation and multi-scale feature fusion; ResNet enabled deeper and more stable learning via residual mappings; ResNeXt and SE-Net refined this paradigm by improving representational diversity and channel adaptivity; and UResNet unified these ideas into a hybrid framework that combines physics-aware preprocessing with deep residual refinement. Together, these developments have significantly advanced the quality, efficiency, and reliability of underwater image enhancement systems.

However, as underwater environments present complex, dynamic, and often unpredictable degradation patterns, CNN-based architectures—though powerful—are limited by their locality and deterministic nature. In principle, one might attempt to overcome these limitations by collecting vast and diverse training datasets so that the network can learn to handle variations in illumination dynamics, turbidity, water color, and optical distortion. In practice, however, this approach is infeasible: acquiring and labeling large-scale underwater datasets across multiple environmental conditions is logistically expensive and physically constrained, while the inherent diversity of underwater scenes cannot be fully captured through data sampling alone. Consequently, improving model generalization through architecture design and hybrid learning strategies remains a more realistic path forward.

It is also worth clarifying that attention mechanisms—often associated with Transformer models—were first introduced and widely applied within CNN architectures. Channel attention (as in SE-Net), spatial attention, and hybrid forms such as CBAM enable convolutional networks to adaptively weight salient features, thereby enhancing their representational flexibility without abandoning the convolutional paradigm. These attention-augmented CNNs effectively bridge the gap between local feature extraction and global context reasoning, laying the conceptual foundation for the later evolution toward Transformer-based and diffusion-inspired models.

The next generation of enhancement frameworks is thus moving toward models that capture global dependencies, probabilistic uncertainty, and data-driven diffusion processes. These directions promise a new wave

of underwater image enhancement research that transcends the representational boundaries of traditional convolutional neural networks.

2.3 U-Net and Its Derivatives

2.3.1 Classic U-Net and Its Relevance

Originally developed for biomedical image segmentation, the U-Net [1] architecture has become a cornerstone in underwater image enhancement due to its strong ability to preserve spatial structure while reconstructing degraded visual content. The U-shaped design, consisting of a contracting path that encodes contextual features and an expanding path that restores spatial resolution, is particularly well-suited for compensating the local contrast and texture loss caused by underwater scattering. Skip connections between symmetric encoder and decoder layers enable direct information flow from low-level features (edges, colors) to high-level semantic representations, preventing the loss of fine details during down sampling.

In underwater enhancement tasks, the U-Net framework provides a flexible basis upon which domain-specific modules can be added [10]. These include multi-scale feature extraction units for handling variations in object scale and illumination, as well as specialized color correction components designed to mitigate the wavelength-dependent color bias. The modularity of U-Net makes it ideal for further hybridization with attention mechanisms, adversarial training, and semi-supervised strategies.

2.3.2 Semi-Supervised U-Net for Underwater Enhancement

One of the key difficulties in underwater image enhancement is the scarcity of paired datasets—degraded underwater images with corresponding clean ground-truth references. To overcome this, researchers have explored semi-supervised U-Net [10] variants that can learn from both labeled and unlabeled data. An improved semi-supervised learning-based U-Net introduces several important modifications to the standard structure.

First, a multi-scale input-level fusion module integrates spatial information from raw underwater images, estimated natural light fields, and optional depth maps. This design allows the network to learn context-aware mappings that capture both local textures and global illumination variations. The use of multi-scale fusion enhances the network’s capacity

to reason about depth-dependent degradation, an essential feature for realistic underwater correction.

Second, the model incorporates a multi-branch hybrid convolutional attention residual block. This block enhances feature extraction by simultaneously emphasizing critical channel information (color components) and spatial regions (edges and objects). By combining different convolutional kernel sizes within each branch, the model effectively captures both fine-grained and global patterns of degradation. Residual connections further stabilize the training process and improve convergence.

Finally, the semi-supervised learning framework itself plays a vital role. It leverages unpaired or partially labeled datasets by introducing consistency losses or pseudo-label mechanisms that regularize the model on unlabeled samples. This not only increases the effective size of the training set but also improves the generalization of the U-Net to unseen underwater domains. Qualitative and quantitative experiments demonstrate that such architectures produce more natural color tones, sharper boundaries, and higher perceptual similarity metrics across diverse underwater scenes.

2.3.3 Dual U-Net for Multi-Stage Enhancement

While a single U-Net can effectively restore color and contrast, its ability to handle severe scattering or light attenuation remains limited. To address this, the Dual U-Net [11] architecture was proposed, consisting of two interconnected U-Nets that operate in sequence. The first U-Net estimates transmittance or intermediate enhancement information from a pre-processed grayscale version of the input image, often augmented with an attention mechanism to highlight key structural regions. The second U-Net then uses this intermediate representation as guidance for full-color enhancement, effectively combining physical modeling with deep feature learning.

The design principle behind Dual U-Net is multi-stage refinement: the first stage focuses on estimating the illumination and transmission properties of the scene, while the second focuses on reconstructing fine textures and color balance. This architecture mirrors the two-step process of traditional model-based underwater restoration (transmission estimation followed by radiance reconstruction) but learns both steps in a data-driven manner. To mitigate the problem of dataset scarcity, training pairs are often generated using unpaired image translation methods such

as CycleGAN, which synthesize degraded images from clean references. Experimental results show that Dual U-Net achieves superior results compared with earlier CNN-based methods, delivering improved structural similarity and color accuracy in both synthetic and real underwater environments.

2.3.4 U-Net Denoising Autoencoders

Another U-Net-inspired direction integrates denoising capabilities into the enhancement process. The Underwater Denoising Autoencoder [1] (UDAE) applies a U-Net architecture to simultaneously remove noise and restore color. Unlike GAN-based methods, which often introduce instability or over-saturation artifacts, UDAE leverages an end-to-end autoencoder framework that directly reconstructs clean images from degraded inputs. Its strength lies in balancing accuracy and computational cost, making it feasible for real-time operation on underwater robots.

The UDAE model processes video frames captured by ROVs or AUVs, reconstructing them to improve perceptual clarity for both human operators and automated visual tasks. By training on synthetically paired datasets, the network learns to recover true colors and reduce blurring without relying on adversarial loss. Comparative experiments reveal that UDAE achieves comparable or superior results to GAN-based color correction models, demonstrating that well-structured convolutional autoencoders can deliver robust enhancement with reduced complexity.

2.3.5 UResNet: Integrating U-Net and Residual Learning

The Underwater ResNet [9] (UResNet) architecture represents the convergence of U-shaped encoder-decoder design and residual learning principles. It inherits the structural advantages of U-Net—symmetrical encoding and decoding with skip connections—and augments them with residual blocks that improve feature propagation and gradient stability. This hybrid design addresses two critical limitations of standard U-Net architectures: the loss of contextual information in deep layers and the difficulty of training very deep networks for complex underwater scenes.

In UResNet, each convolutional stage within the encoder and decoder is replaced by a residual block, allowing features to be refined through identity-based shortcuts. These residual pathways ensure that both low-level color cues and high-level contextual patterns are effectively preserved. As a result, UResNet exhibits enhanced robustness to the

diverse degradations encountered in underwater imagery, such as color casts, contrast loss, and scattering blur.

A major contribution of recent research on UResNet is the combination of hybrid preprocessing pipelines with structure-aware residual learning. In one such approach, the enhancement process begins with a hybrid preprocessing stage that applies Underwater White Balance (UWB) and Variational Contrast and Saturation Enhancement (VCSE). The UWB component compensates for color bias by adjusting the relative intensities of red, green, and blue channels based on estimated ambient illumination, while VCSE enhances local contrast and saturation through an energy-variational optimization model. The preprocessed image is then fed into the UResNet, where residual learning refines the enhancement by recovering fine structures and edges lost during preprocessing.

Three UResNet variants have been explored: the standard baseline, a Sobel-filter-augmented version that emphasizes edge information, and an SE-integrated version that introduces adaptive channel weighting. The SE-UResNet variant consistently achieves the best performance, delivering higher color fidelity, improved contrast, and sharper details across multiple underwater datasets such as EUVP. This demonstrates that coupling data-driven residual enhancement with physics-inspired preprocessing yields superior results compared to purely data-driven models. The hybrid design ensures that color correction and texture restoration are handled in complementary domains—the preprocessing stage corrects global illumination and color bias, while the deep model enhances local details and perceptual realism.

An important insight from UResNet studies is that architecture alone is insufficient to guarantee robust enhancement; the data pipeline and prior modeling are equally critical. By embedding optical priors into the preprocessing step and learning residual corrections in the deep network, UResNet strikes a balance between physical interpretability and learning capacity. This balance allows it to generalize effectively across diverse underwater scenes, from shallow coastal waters to deep-sea environments.

2.4 GAN and Diffusion-based Approaches for Underwater Image Enhancement

With the growing limitations of traditional convolutional networks in capturing global illumination shifts and color attenuation inherent in underwater scenes, generative adversarial networks (GANs) have emerged as a transformative framework for underwater image

enhancement (UIE). Unlike conventional feed-forward models that rely on pixel-level losses, GAN-based approaches leverage an adversarial training paradigm to learn complex data distributions, enabling more realistic texture recovery, perceptual color correction, and content-aware enhancement. Over time, this paradigm evolved from early unsupervised generation frameworks like WaterGAN[12] to physically informed models such as PUGAN [13], and eventually to hybrid and diffusion-based paradigms such as DiffUIE [14] and WaterDiffusion [15], which incorporate probabilistic priors and iterative refinement for both visual restoration and semantic understanding.

Early efforts such as WaterGAN marked a key milestone by introducing an unsupervised generative pipeline that simulated underwater image formation from in-air image and depth pairs. This work recognized a fundamental bottleneck in underwater vision—the lack of large-scale paired datasets. Rather than requiring ground truth clean images captured in controlled underwater environments, WaterGAN synthesized realistic underwater imagery based on a learned mapping that mimics light absorption and scattering through the water column. This synthetic data generation allowed the authors to train a downstream correction network capable of real-time monocular color restoration, bridging the gap between theoretical imaging models and practical deep learning pipelines. The unsupervised learning setup was particularly significant because it reduced the reliance on costly calibration and manual data annotation, establishing a new direction for domain adaptation in underwater enhancement.

Building on this, subsequent approaches began to emphasize visual quality and perceptual fidelity. The model proposed in Fast Underwater Image Enhancement [16] for Improved Visual Perception introduced a conditional GAN (cGAN) framework tailored for real-time enhancement with explicit perceptual supervision. The authors formulated a composite loss function combining global content, local texture, and style information to better guide adversarial learning. More importantly, they introduced EUVP, one of the earliest large-scale paired and unpaired underwater image datasets, enabling the network to learn enhancement mappings under a wide range of water types and visibility conditions. This dataset-driven improvement enabled consistent enhancement results that benefited downstream perception tasks, such as underwater object detection and human pose estimation, demonstrating that perceptual enhancement directly contributes to functional improvements in vision-based robotics. The explicit inclusion of style loss and multi-domain supervision further refined the GAN framework’s ability to maintain

natural color balance while avoiding over-saturation, a common artifact in earlier adversarial approaches.

To further bridge the gap between perceptual quality and task-specific performance, Twin Adversarial Contrastive Learning for Underwater Image Enhancement [17] and Beyond introduced an object-guided twin adversarial architecture. This work proposed a closed-loop adversarial system coupled with contrastive learning, designed not only to enhance visual appeal but also to improve object detection accuracy in degraded underwater scenes. The bilateral adversarial structure incorporated inverse mappings, effectively learning both degradation and restoration processes within a unified framework. Additionally, the inclusion of a task-aware feedback module provided a novel bridge between low-level image enhancement and high-level vision tasks, allowing gradients from detection networks to guide enhancement toward feature-preserving transformations. This represented a major conceptual advance: underwater image enhancement was no longer seen as an isolated preprocessing step, but as an integrated component of perception systems for autonomous underwater vehicles. The framework achieved superior visual quality metrics (UIQM, SSIM, and PSNR) and significant gains in downstream object detection performance, confirming the viability of adversarial-contrastive joint optimization.

While these advances demonstrated the flexibility of adversarial learning, pure GAN models often suffer from instability in training, mode collapse, and inconsistency in physical realism. To mitigate these issues, hybrid models combining physical priors with adversarial frameworks began to appear. The PUGAN model exemplifies this integration by incorporating underwater imaging physics directly into the GAN training process through a dual-discriminator structure. One discriminator ensured visual plausibility, while the other enforced physical consistency with the underwater degradation model, such as wavelength-dependent attenuation. By aligning adversarial learning with the physical characteristics of light propagation, PUGAN achieved superior adaptability across diverse water types and illumination conditions. This hybridization of learning-based and physics-based approaches significantly improved model generalization, ensuring that the enhancement remained consistent even in highly turbid or color-dominant environments.

While GAN-based frameworks established a solid foundation for underwater image enhancement by learning realistic mappings between degraded and high-quality domains, they also revealed several inherent

limitations. Chief among these were unstable adversarial training, mode collapse, and the tendency to overfit to perceptual metrics rather than physical fidelity. Furthermore, despite improved realism, many GAN-enhanced results exhibited over-saturated colors or nonuniform brightness—issues stemming from the generator’s lack of explicit probabilistic understanding of image formation. To overcome these constraints, researchers have increasingly turned to diffusion models, which introduce a fundamentally different generative paradigm based on iterative denoising and probabilistic modeling of image priors.

Diffusion-based models, unlike GANs, do not rely on an adversarial discriminator; instead, they learn to progressively reconstruct a clean image from noise through a sequence of denoising steps. This allows them to capture global statistical structure while maintaining training stability and diversity in generation. In the context of underwater imagery, where degradation stems from complex interactions of scattering, absorption, and refraction, the capacity of diffusion models to represent multi-modal distributions and learn detailed priors proves particularly advantageous.

One of the earliest dedicated diffusion-based UIE methods, UDBE [18] (Unsupervised Diffusion-based Brightness Enhancement), specifically targeted the brightness degradation problem often neglected in traditional approaches. The authors observed that underwater scenes captured at greater depths suffer not only from color shift but also from significant luminance attenuation, producing visually dark and detail-sparse images. UDBE addressed this challenge through a conditional diffusion framework that combines the input image with a color map and a signal-to-noise ratio (SNR) map to guide the denoising process. This conditioning mechanism ensured that the diffusion process enhanced brightness without distorting chromatic balance. The unsupervised nature of UDBE made it well suited for real-world underwater conditions where clean reference images are unavailable. Experiments across benchmark datasets (UIEB, SUIM, RUIE) showed substantial gains in both quantitative metrics (PSNR, SSIM, UIQM) and perceptual visual quality, underscoring the robustness and interpretability of diffusion-based brightness modeling compared to conventional GAN architectures.

Following UDBE, diffusion models were expanded beyond brightness enhancement to full-spectrum underwater image restoration. The DiffUIE model represented a key step in this direction, introducing a latent global prior-guided diffusion network designed specifically for UIE. Rather than treating enhancement as a purely visual reconstruction task, DiffUIE incorporated a global feature prior that captures long-range dependencies

and contextual relationships between color channels. By embedding an explicit underwater degradation model into the training process, DiffUIE learned to map degraded images to their restored counterparts through iterative refinement guided by physical realism. The introduction of frequency modulation scaling further prevented the over-enhancement of high-frequency features—a common artifact in deep diffusion networks. Additionally, the authors employed a backward diffusion process to stabilize inference, reducing artifacts during image reconstruction. Evaluations across multiple public datasets demonstrated that DiffUIE consistently outperformed GAN-based methods and even surpassed conventional CNN-based approaches in both subjective assessments and objective measures such as UIQM, UCIQE, and SSIM.

Beyond single-task enhancement, diffusion models have also been leveraged for multi-objective underwater vision, exemplified by WaterDiffusion, a prior-involved unrolling diffusion model for joint saliency detection and visual restoration. This work recognized that underwater image degradation not only affects human perception but also impairs high-level vision tasks such as object localization and semantic segmentation. Instead of performing enhancement and detection sequentially, WaterDiffusion integrated both into a single collaborative framework. The model introduced a self-attention joint diffusion mechanism, embedding dynamically updated saliency masks into the denoising process to guide feature emphasis toward salient regions. Moreover, by extending the underwater degradation prior into a multiscale decoder, the model exploited optical transmission maps to assist in accurate localization of salient objects. The introduction of a gate-guided binary indicator allowed adaptive channel selection, further improving generalization across different water types. Through an unrolled optimization process based on Half-Quadratic Splitting, WaterDiffusion refined saliency masks iteratively, achieving superior quantitative and qualitative performance compared to existing SOD (salient object detection) and UIE baselines. This coupling of enhancement and detection tasks within a diffusion framework signified a conceptual shift from passive restoration to task-aware generative modeling.

An interesting parallel development is the use of diffusion for data generation in underwater perception. In the study Side-Scan Sonar Underwater Target Detection [19]: Combining the Diffusion Model With an Improved YOLOv7 Model, diffusion models were employed not for enhancement per se, but for augmenting limited sonar imagery datasets. By generating denoised and visually consistent synthetic sonar images, the authors significantly improved the training diversity and robustness of an

object detector built on YOLOv7. This demonstrates a secondary but strategically important role of diffusion models in the underwater domain: they can serve as data synthesizers to overcome the persistent scarcity of labeled underwater imagery, thereby indirectly contributing to enhancement and detection performance.

When comparing diffusion-based UIE approaches with their GAN counterparts, several conceptual distinctions emerge. First, diffusion models inherently model probability distributions over the data manifold, which allows them to reconstruct natural-looking images without relying on adversarial loss functions. This leads to more stable convergence and avoids the discriminator-generator imbalance characteristic of GAN training. Second, diffusion methods possess a controllable denoising trajectory, enabling fine-grained manipulation of enhancement strength, brightness, and detail preservation at inference time. Third, unlike GANs, which often learn implicit mappings prone to overfitting, diffusion frameworks can explicitly incorporate domain priors, such as physical light propagation models or saliency constraints, directly into the sampling process. These properties collectively result in enhanced interpretability, robustness to domain shifts, and superior generalization across diverse underwater conditions.

The recent diffusion-based models also signal a broader methodological transition in underwater image enhancement research: from purely discriminative enhancement networks to probabilistic generative systems that unify restoration, understanding, and data synthesis. In this emerging paradigm, enhancement is no longer an end in itself but part of a larger pipeline supporting perception, navigation, and object recognition in marine robotics. Furthermore, diffusion models offer natural compatibility with reinforcement learning and transformer-based architectures, suggesting future directions where underwater vision systems may achieve self-adaptive enhancement conditioned on task objectives or environmental cues.

In summary, the evolution from GAN to diffusion-based UIE reflects an increasing emphasis on stability, interpretability, and multi-task integration. Early GANs such as WaterGAN laid the groundwork for unsupervised realism, followed by perceptual and task-oriented adversarial designs. Hybrid models like PUGAN introduced physical awareness, while diffusion frameworks such as UDBE, DiffUIE, and WaterDiffusion redefined enhancement as a probabilistic denoising process grounded in domain priors. Together, these works mark a decisive

shift toward physics-informed, probabilistic generative modeling as the future trajectory of underwater image enhancement research.

2.5 Evaluation Metrics for Underwater Image Enhancement

Evaluating underwater image enhancement (UIE) objectively remains a persistent challenge due to the complex and variable nature of underwater degradation. Traditional image quality metrics such as PSNR and SSIM, though widely adopted in general computer vision tasks, fail to capture perceptual color fidelity and structural consistency under the wavelength-dependent attenuation and scattering unique to underwater environments. To address this gap, a specialized metric known as the Underwater Color Image Quality Evaluation (UCIQE) was proposed as a domain-specific quantitative standard for assessing the visual quality of enhanced underwater images.

The UCIQE metric [20] was derived through extensive subjective experiments, analyzing human perceptual judgments of underwater image quality across diverse environmental conditions. By examining the statistical distribution of image pixels in the CIELab color space, researchers identified three primary components that strongly correlate with human perception: chroma, saturation, and contrast. These factors collectively capture the color distortion, blurriness, and low-contrast phenomena that dominate underwater imagery. UCIQE models these as a linear combination of the three components, effectively quantifying the degree of nonuniform color cast, haze, and contrast degradation in a single, interpretable scalar score.

Experimental validation demonstrated that UCIQE performs competitively against general-purpose natural image quality metrics, while showing higher correlation with subjective Mean Opinion Scores (MOS) on underwater datasets. Its computational simplicity and robustness make it suitable not only for post-processing evaluation but also for real-time assessment in underwater robotic or monitoring systems. Moreover, UCIQE's independence from reference images allows it to serve as an effective metric for evaluating unsupervised and physics-based enhancement methods, where ground-truth clean images are unavailable.

In summary, UCIQE represents a critical step toward task-relevant, perception-aligned evaluation in underwater imaging research. By bridging the gap between human visual perception and computational measurement, it provides a consistent and efficient benchmark for

comparing both traditional and learning-based UIE algorithms, and remains one of the most widely adopted quantitative standards in the field.

3 Methodology

This section outlines the overall methodological pipeline adopted in this study. First, the characteristics of the input data are presented, including its origin, structure, and the preprocessing steps required to ensure consistency during training and inference. Next, the machine learning component used in the system is introduced, with an explanation of why this model was selected over alternative architectures and how it integrates with the broader enhancement workflow. Building upon the model’s output, a novel computer vision–based post-processing algorithm is described, designed to refine visual quality and address artifacts that the learning framework alone cannot resolve. Finally, the evaluation strategy is detailed, covering both the quantitative metrics and the qualitative assessments employed to measure performance and validate the effectiveness of the proposed approach.

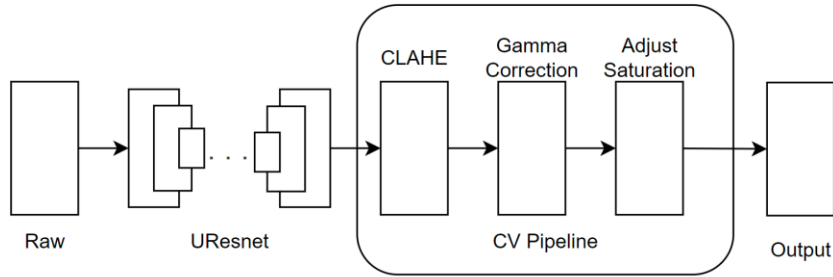


Figure 1: UResnet and CV Pipeline structural

3.1 Datasets

3.1.1 UIEB

The Underwater Image Enhancement Benchmark (UIEB) [21] dataset is one of the most widely used benchmarks for evaluating underwater image enhancement algorithms. It contains 950 underwater images, divided into two subsets: 890 raw underwater images with corresponding high-quality reference images, and 60 challenging underwater images without reference data. The dataset was designed to provide a comprehensive evaluation platform that reflects the complexity of real underwater environments,

including color distortion, low contrast, and spatial non-uniformity caused by light scattering and absorption.

UIEB was originally introduced in Water-Net [22] and has since become a standard dataset in underwater image enhancement research. The paired ground truth images were generated through a semi-manual selection and refinement process, ensuring perceptual fidelity and consistent visual quality. Researchers are encouraged to report their enhancement results on this dataset, as the maintainers periodically update the public leaderboard to include new methods.

3.1.2 LSUI

The Large-Scale Underwater Image (LSUI) [23] dataset was developed to overcome the limitations of previous underwater datasets in terms of scale, scene diversity, and annotation richness. It comprises 4,279 real-world underwater image groups, each containing a raw underwater image paired with a high-fidelity reference image, a semantic segmentation map, and a medium transmission map. Compared with earlier datasets such as UIEB, LSUI includes a broader range of underwater scenes and provides higher-quality references for data-driven methods.

The LSUI dataset was introduced alongside a U-shape Transformer model for underwater image enhancement, which integrates channel-wise and spatial-wise attention mechanisms to address non-uniform color attenuation.

3.2 Models

3.2.1 Overview of Model Architecture

In this work, we propose a lightweight yet effective underwater image enhancement network named UResNet-P, designed to restore natural color tones, improve contrast, and preserve fine structural details in degraded underwater images. The model is inspired by the U-ResNet architecture but simplified and optimized for practical underwater scenarios. It integrates three core principles: residual feature learning, hierarchical convolutional representation, and edge-aware enhancement.

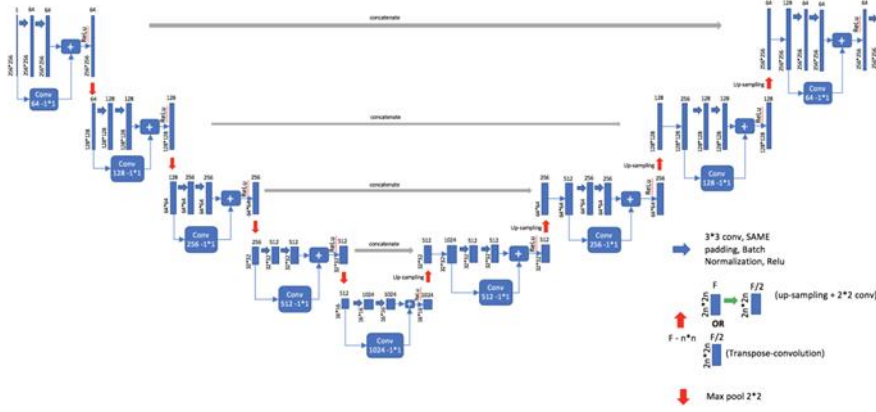


Figure 2: UResNet Model

UResNet-P is built around a stack of residual blocks that focus on learning the difference between degraded input and the clean target image, rather than performing direct pixel-wise mapping. This residual design helps the model converge faster, prevents gradient vanishing, and enables deeper feature representation without significant computational overhead. Additionally, an auxiliary edge detection branch is introduced to explicitly guide the network toward maintaining sharp boundaries, thus improving perceptual quality and image realism.

Building upon the original U-ResNet architecture, this work introduces several structural refinements aimed at improving the network's robustness in diverse underwater conditions. Most notably, the residual blocks have been redesigned to strengthen feature extraction while maintaining computational efficiency. Instead of the standard two-layer residual structure, each modified block incorporates enhanced convolutional pathways that facilitate deeper receptive fields and more stable gradient propagation. In addition, a Squeeze-and-Excitation (SE) module is appended to the output of every residual block to provide channel-wise attention. This mechanism enables the network to dynamically recalibrate feature responses by emphasizing informative channels and suppressing less relevant ones. Through this integration of attention-augmented residual learning, the model gains improved generalization when handling variations in water quality, lighting absorption, and color distortion, ultimately producing more consistent enhancement results across different underwater scenes.

3.2.2 Network Components and Design Motivation

The UResNet-P model consists of three major components: an input/output convolutional layer, a multi-layer residual block stack, and an edge detection branch.

The input convolution transforms the 3-channel RGB input into 64 feature maps using a

3×3 convolution kernel with stride 1 and padding 1. This step captures the fundamental local structures, such as color gradients and low-level textures. The extracted feature representation then passes through a sequence of 16 residual blocks, each containing two convolutional layers followed by batch normalization and a ReLU activation function. A skip connection adds the input feature map to the block’s output, enabling the network to efficiently learn residual mappings.

Each residual block helps the model refine image features progressively, with early layers correcting coarse color bias and deeper layers focusing on texture restoration and contrast enhancement. Finally, an output convolution layer converts the 64 feature maps back to a 3-channel RGB image, generating the enhanced output.

3.2.3 Edge-Aware Enhancement Mechanism

To strengthen spatial sharpness and prevent over-smoothing—an issue often seen in deep enhancement models—UResNet-P incorporates an explicit Edge_Detector module. This branch applies a fixed Laplacian-like convolution kernel that highlights high-frequency areas, such as object contours and texture transitions.

By computing an edge map from the enhanced output, the network gains additional structural feedback during training. Although this edge information is not directly used in the loss function in the current configuration, it provides a meaningful intermediate representation for visual validation and could be extended in future work to form an edge-guided loss. This design encourages the model to pay more attention to local geometry and helps maintain realistic structural integrity, which is critical for underwater scenes with complex illumination patterns.

3.2.4 Loss Function Design

The model is trained under a fully supervised paradigm using the Mean Squared Error (MSE) loss function:

$$L_{MES} = \frac{1}{N} \sum_{i=1}^N (I_{pred}^{(i)} - I_{gt}^{(i)})^2$$

Where I_{pred} denotes the enhanced image generated by the network, and I_{gt} is the corresponding ground truth image. The MSE loss minimizes the pixel-wise intensity difference between prediction and reference, enforcing global color consistency and smooth reconstruction.

The use of MSE is particularly suitable for underwater enhancement, as it provides a stable and interpretable gradient for regression tasks. While perceptual or adversarial losses could further refine texture realism, MSE ensures robust convergence and avoids introducing unstable artifacts. This makes it an appropriate baseline for validating the network’s architecture effectiveness.

3.2.5 Training Strategy and Implementation Details

The model was trained using the AdamW optimizer, which combines Adam’s adaptive learning rate mechanism with decoupled weight decay regularization. This choice helps maintain training stability while preventing overfitting, which is especially important when the dataset includes diverse underwater conditions. The weight decay coefficient was set to 1×10^{-4} , encouraging sparsity in model parameters and improving generalization.

The learning rate was initialized as *config.lr* and dynamically adjusted during training using the ReduceLROnPlateau scheduler. This scheduler monitors the validation loss and reduces the learning rate by a factor of *config.decay_rate* when no improvement is observed for 10 consecutive epochs. This adaptive strategy prevents the optimizer from getting stuck in suboptimal minima and allows finer convergence as training progresses.

The data preprocessing pipeline includes resizing all input and target images to a fixed resolution (*config.resize* $\times$ *config.resize*) and converting them to normalized tensors. Each batch of input-target pairs is loaded using PyTorch’s DataLoader with shuffling enabled to improve gradient diversity.

During each epoch, the model is trained in mini-batches. For each batch, the input image is passed through the network to generate an enhanced output and its corresponding edge map. The pixel-wise MSE between the predicted and ground truth images is computed and backpropagated through the network. Model parameters are updated using the AdamW optimizer, and both training and validation losses are recorded for monitoring convergence.

Validation is performed at the end of every epoch in evaluation mode (*model.eval()*), ensuring that batch normalization and dropout behave consistently. The validation loss serves as a criterion for learning rate adjustment and model checkpoint saving. Periodically, the model weights are stored as .ckpt files in the snapshot directory, enabling recovery and later evaluation.

The logging system records epoch-wise training and validation losses, providing quantitative feedback for assessing model stability and progress. As training proceeds, a decreasing trend in both training and validation losses indicates effective learning and generalization.

3.2.6 Discussion and Summary

The proposed UResNet-P architecture and training framework jointly address the major challenges of underwater image enhancement—namely, color distortion, uneven illumination, and structural blur. The residual block design facilitates efficient deep feature extraction without gradient degradation, while the edge detection branch ensures spatial fidelity.

Using MSE loss and the AdamW optimizer offers a balance between simplicity, convergence stability, and generalization capability. The combination of learning rate scheduling and periodic checkpointing ensures adaptive optimization and prevents overfitting. Despite the simplicity of the loss function, the network achieves high-quality enhancement results, producing images with improved visibility, contrast, and color balance.

In summary, UResNet-P represents a practical and effective model for underwater image enhancement. Its lightweight residual design, explicit edge modeling, and robust training strategy make it suitable for deployment in both research and real-world embedded systems. The proposed framework not only achieves reliable visual restoration but also provides a flexible foundation for future extensions, such as multi-scale fusion, perceptual loss integration, or transformer-based global feature modeling.

3.3 Traditional CV Enhancement Pipeline

Although the proposed UResNet-P network effectively restores underwater image color and contrast, some residual imperfections may remain after the deep enhancement stage. These include mild color imbalance, insufficient contrast in localized regions, or low saturation

caused by the complex light attenuation characteristics of underwater environments. To further refine the visual quality and produce more perceptually pleasing results, we designed a traditional computer vision-based post-processing pipeline that operates on the enhanced outputs of the neural network.

This pipeline integrates several classical image enhancement techniques—white patch balance, contrast-limited adaptive histogram equalization (CLAHE), gamma correction, and saturation adjustment—into a flexible and modular framework. Each operation targets a specific visual property, and their combination helps compensate for the limitations of purely data-driven enhancement.

3.3.1 White Patch Balance.

The white patch balance method assumes that the brightest pixels in an image should be pure white. By scaling each color channel according to a specified percentile (e.g., 95th percentile), this step corrects global color bias and eliminates the blue-green cast commonly found in underwater imagery. This operation enhances color fidelity while preserving highlights and avoids overexposure.

3.3.2 Adaptive Histogram Equalization (CLAHE).

To improve local contrast, the image is converted to the LAB color space, and the luminance channel (L) undergoes CLAHE. Unlike global histogram equalization, CLAHE enhances contrast within small spatial tiles, limiting the amplification of noise through a clipping threshold. This results in a balanced brightness distribution, making objects more distinguishable even in regions with uneven illumination.

3.3.3 Gamma Correction.

Underwater scenes often appear dim due to light absorption in deeper water. Gamma correction adjusts the overall brightness and mid-tone contrast by applying a nonlinear intensity transformation

$$I_{out} = I_{in}^{1/\gamma}$$

By setting $\gamma \approx 1.1 \sim 1.2$, the output image becomes visually brighter and more natural without over-saturating bright regions.

3.3.4 Saturation Adjustment.

The final step involves adjusting image saturation in the HSV color space. The saturation channel is scaled by a small factor (e.g., 1.1), which strengthens color vividness and improves perceptual aesthetics. This operation is particularly effective for underwater scenes, where excessive scattering tends to wash out chromatic contrast.

Four methods were implemented—White Patch Balance, CLAHE, Gamma Correction, and Saturation Adjustment—with various combinations evaluated across multiple metrics. By comparing both the mean and median performance of each configuration, the pipeline using CLAHE + Gamma Correction + Saturation Adjustment was selected as the optimal combination, providing more stable and visually natural enhancement results.

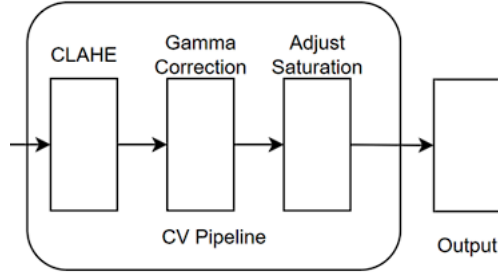


Figure 3: CV Pipeline Design

This modularity enables users to easily experiment with different combinations or parameter settings depending on image characteristics. Batch processing is supported for large-scale datasets, and enhanced images are automatically saved for evaluation.

In essence, this hybrid approach combines the global semantic restoration capability of deep learning with the fine-grained tonal adjustments of traditional image processing. By applying this pipeline to the outputs of UResNet-P, we achieved clearer textures, more balanced illumination, and richer color representation. The combination of neural enhancement and classical post-processing therefore provides a robust, interpretable, and computationally efficient solution for underwater image enhancement.

3.4 Evaluation Framework

To ensure a fair comparison, the proposed deep learning model was trained and evaluated under the same experimental conditions as the baseline method, Diffwater. Both models were trained using the LSUI

training subset and tested on the validation and test subsets without any data overlap. The training was performed using identical image resolutions, data augmentations, and loss configurations, so that any performance differences could be attributed solely to the algorithmic design.

The evaluation considered three widely adopted quantitative metrics for underwater image enhancement:

Peak Signal-to-Noise Ratio (PSNR), which measures reconstruction fidelity and overall image quality.

Structural Similarity Index (SSIM), which reflects the preservation of structural and textural details.

Underwater Color Image Quality Evaluation (UCIQE), a non-reference metric that quantifies the perceptual quality of color and contrast in underwater scenes.

These indicators jointly reflect both the physical correctness and perceptual realism of enhanced images.

4 Results

In this section, The experiments were designed to assess both the quantitative performance and the qualitative visual quality of the results, as well as to validate the effectiveness of the additional traditional computer vision (CV) processing pipeline. All experiments were conducted on the LSUI dataset, which contains diverse underwater scenes with varying lighting, turbidity, and color distortion conditions.

4.1 Qualitative Evaluation

Beyond numerical evaluation, a direct visual comparison was performed to qualitatively assess perceptual quality. The figure below shows sample outputs from both models given the same input images. The Diffwater model tends to produce results with residual haze and a greenish tone, while our approach recovers more balanced colors and sharper edges. Moreover, the proposed method effectively suppresses backscatter effects and enhances local contrast, leading to clearer object boundaries and more visually pleasing scenes.

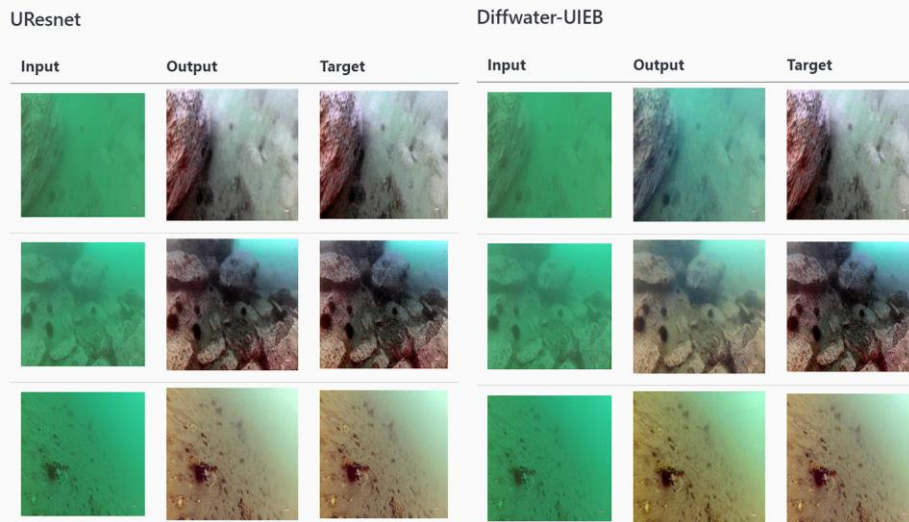


Figure 4: UResNet vs Diffwater

These qualitative observations are consistent with the quantitative findings and further support the superiority of the proposed network in restoring both global color fidelity and fine local structures.

4.2 Quantitative Evaluation

Table 1 presents the comparison results between the proposed model and Diffwater on the LSUI dataset. As shown, our model achieves consistently higher scores across all three metrics. In particular, the PSNR and SSIM improvements demonstrate the model’s stronger ability to restore fine-grained structures, while the higher UCIQE values indicate a more natural color balance and better contrast in the enhanced results.

<i>Metrics</i>	<i>UResnet</i>	<i>Diffwater</i>
<i>PSNR(dB) Avg.</i>	31.68	20.89
<i>UCIQE Avg.</i>	103.1682	106.7373
<i>SSIM Avg.</i>	0.915	0.8406

Table 1 Quantitative comparison of PSNR, SSIM, and UCIQE on LSUI dataset

4.3 Integration with CV-Based Post-Processing

To further enhance the final image quality, we integrated the trained UResNet model with a series of traditional computer vision (CV) post-processing pipelines. The purpose of this experiment is to analyze how classical enhancement operations can complement deep learning outputs. The CV pipeline includes adjustable combinations of white balance correction, histogram equalization, guided filtering, and contrast stretching.

Each variant of the pipeline was applied to the outputs of the UResNet model, and the results were quantitatively assessed using the UCIQE metric, as it effectively captures perceptual improvements that may not be reflected in PSNR or SSIM. The combination achieving the highest UCIQE score was selected as the optimal pipeline configuration.

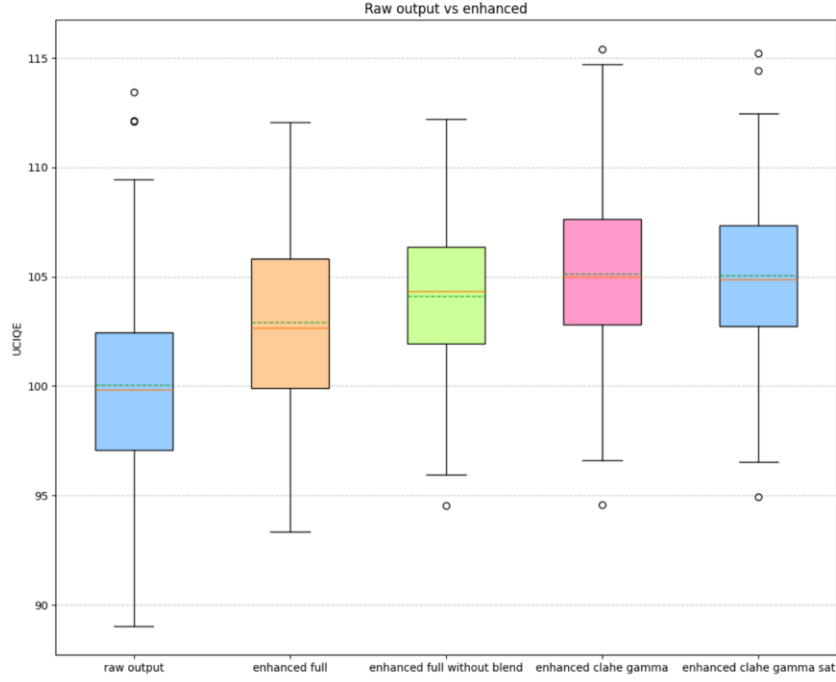


Figure 5: UCIQE Metrics with different CV Pipeline

The experiments reveal that the integration of lightweight CV operations can further refine the visual appearance of the enhanced images, particularly in terms of global color consistency and local detail sharpness. This demonstrates the complementary nature of deep learning-based and traditional image processing techniques in underwater image enhancement.

4.4 Summary of Findings

Overall, the experimental results validate the effectiveness of the proposed method. Compared with the Diffwater baseline, our model delivers superior quantitative performance and visually more natural results. Furthermore, by incorporating an optimized CV post-processing pipeline, the final enhancement quality is further improved, achieving the best balance between structural preservation, color correction, and perceptual realism.

These results highlight the potential of combining learning-based and rule-based techniques to address the challenges of underwater image enhancement in complex and variable real-world conditions.

5 Discussion

The experimental results demonstrate that adding the classical CV pipeline offers clear benefits to the overall enhancement quality. In particular, the combination of CLAHE, Gamma Correction, and Saturation Adjustment provides roughly a 5% improvement in the UCIQE score over the raw network output. This indicates that while the deep model restores global structures and basic color balance, the CV stage effectively fine-tunes contrast and channel relationships, producing more stable and natural-looking results.

In terms of sample-specific behavior, images with moderately bluish tones and adequate illumination achieved the best enhancements. These inputs fall within the model’s preferred distribution, allowing it to correct color absorption and improve contrast without significantly amplifying noise. Conversely, images with severe color cast or extremely low brightness exhibit inferior results. A plausible explanation is that when the input deviates too far from typical underwater lighting and absorption characteristics, the model has difficulty estimating reliable color priors, and the CV pipeline may over-amplify noise or exaggerate incorrect color channels.

Despite the improvements, several limitations are present. The current architecture is not optimized for heavy turbidity or strong backscatter, resulting in suboptimal recovery in highly degraded waters. The model’s performance also depends strongly on the range of its training data. Moreover, the added CV operations, although useful on average, do not adapt dynamically to scene conditions. Another notable limitation is that the model occasionally over-enhances high-frequency edges, which can unintentionally amplify noise or create halo-like artifacts around object boundaries, especially in low-light scenarios.

An interesting observation is that, in some moderately degraded images, the CV pipeline reveals fine textures that were not visible in the model output alone. This opens the possibility for future hybrid methods

where the network selectively controls CV operators based on learned scene characteristics.

6. Conclusion

This work presented a comprehensive investigation into several mainstream underwater image enhancement methods and proposed a hybrid framework that integrates a UResNet-based deep learning model with a carefully designed classical computer vision pipeline. Through systematic evaluation on the LSUI dataset, the experimental results demonstrated an average improvement of approximately 5% in the UCIQE metric, indicating more balanced color reproduction, improved contrast, and enhanced perceptual clarity compared with the standalone model output and existing approaches.

Several key insights were identified. Deep models such as UResNet effectively capture nonlinear mappings between degraded and reference images, but their outputs may still exhibit mild color imbalance or local artifacts. The addition of lightweight CV post-processing refines global appearance and supplies interpretable adjustments, forming a practical balance between automation and controllability. Importantly, the findings show that learning-based models and traditional enhancement methods are complementary rather than competing, enabling more robust performance across diverse underwater environments.

7. Future Work

A number of promising research directions can be pursued to extend this work. First, deploying the proposed framework on lightweight or resource-constrained platforms—such as underwater robots, ROVs, and embedded vision systems—requires achieving real-time performance under limited computational budgets. This motivates further exploration of model compression, quantization, pruning, and efficient UResNet variants enabled by neural architecture search or knowledge distillation.

Second, enhancing model generalization remains a critical goal. Although the current framework performs well on LSUI, real-world underwater environments vary greatly in depth, turbidity, illumination, and scattering. Future work may incorporate unsupervised domain adaptation, physics-guided synthetic data, or contrastive representation learning to improve robustness. Additional modalities such as depth or polarization images could further support restoration under heavy scattering.

Finally, developing end-to-end low-consumption enhancement systems represents a longer-term vision. Integrating model-level optimization, adaptive CV pipelines, and hardware-aware inference could deliver underwater enhancement that is both computationally efficient and perceptually reliable. Such advancements can directly benefit applications in marine exploration, biological observation, archaeological documentation, and autonomous underwater navigation.

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